

UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Master programme in Data Science and Business Informatics

DATA MINING: FUNDAMENTAL

Project report: Seller-Level Vehicle Listings Dataset

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1 Data Understanding and Preparation

1.1 Data semantics

The Seller-Level Vehicle Listings dataset contains 13,843 records and 33 features. Each record represents a distinct seller (`sp_id`) with information on the seller's vehicle listings. A semantic description table is provided below to get a thorough overview of the dataset.

Table 1.1: Semantic description of the dataset variables. The bold row marks the prediction target.

Feature Name	Description	Datatype	Msrmt Scale
sp_id	Unique identifier of the seller/dealer	float64	Identifier
n_listings	Total number of vehicle listings	int64	Numerical
price_median	Median advertised price of vehicles	float64	Numerical
savings_amount_median	Median advertised savings amount	float64	Numerical
daysonmarket_median	Median days vehicles remained on market	float64	Numerical
mileage_median	Median vehicle mileage	float64	Numerical
year_median	Median vehicle model year	float64	Numerical
vehicle_age_median	Median vehicle age at listing time	float64	Numerical
owner_count_median	Median number of previous vehicle owners	float64	Numerical
horsepower_median	Median vehicle horsepower	float64	Numerical
engine_displacement_median	Median vehicle engine displacement	float64	Numerical
city_fuel_economy_median	Median city fuel economy	float64	Numerical
highway_fuel_economy_median	Median highway fuel economy	float64	Numerical
back_legroom_median	Median rear-seat legroom	float64	Numerical
front_legroom_median	Median front-seat legroom	float64	Numerical
fuel_tank_volume_median	Median fuel tank capacity	float64	Numerical
height_median	Median vehicle height	float64	Numerical
length_median	Median vehicle length	float64	Numerical
width_median	Median vehicle width	float64	Numerical
pct_isCab	Proportion of vehicles previously used as cabs	float64	Numerical
pct_is_new	Proportion of vehicles classified as new	float64	Numerical
pct_salvage	Proportion of vehicles with salvage status	float64	Numerical
pct_theft_title	Proportion of vehicles with theft titles	float64	Numerical
n_listing_days	Distinct calendar days with observed listings	int64	Numerical
listing_span_days	Span in days between earliest and latest listing	int64	Numerical
dominant_body_type	Most frequent vehicle body type	str	Categorical
dominant_fuel_type	Most frequent fuel type	str	Categorical
dominant_transmission	Most frequent transmission type	str	Categorical
dominant_wheel_system	Most frequent drivetrain configuration	str	Categorical
dominant_listing_color	Most frequent exterior color	str	Categorical
franchise_status	Franchise status (franchise, independent, etc.)	str	Categorical
new_used_profile	Inventory composition (new_only, used_only, etc.)	str	Categorical
seller_rating_class	**Discretized target rating (low, medium, high)**	**str**	**Ordinal**

1.2 Distribution of variables and statistics

1.2.1 Summary of statistics

Table 1.2 provides notable descriptions of the primary statistics for each numerical feature. First, the feature magnitudes vary across four orders of scale, from bounded proportions (`pct_*` $\in [0, 1]$) and cabin dimensions (`height_median` mean 65.15 in) to `price_median` (up to $236,438) and `mileage_median` (up to 238,541 mi). Therefore, the data should be standardised before fitting distance-based models.

Secondly, volume, pricing, and risk-history features exhibit severe right-skewness. The most extreme are `pct_theft_title` (14.11), `pct_salvage` (10.34), `daysonmarket_median` (8.75) and `n_listings` (5.25). Conversely, `year_median` (-2.25) and `back_legroom_median` (-3.53) are left-skewed.

Finally, `engine_displacement_median` lacks 3,393 records (24.51%), and sparse risk indicators (`pct_theft_title`, `pct_salvage`) yield IQR $= 0$ due to being heavily zero-inflated.

Table 1.2: Summary statistics for the 24 numerical variables (identifier excluded).

Variable	Count	Mean	Std	Min	25%	50%	75%	Max	Missing	Skew	Kurtosis	IQR
n_listings	13843.0	118.10	137.06	5.00	25.00	70.00	168.00	4959.00	0	**5.25**	120.32	143.00
price_median	13843.0	23106.14	14365.07	1699.00	12495.00	22385.00	29888.00	236438.00	0	**3.45**	34.14	17393.00
savings_amount_median	13843.0	446.21	519.46	0.00	0.00	358.00	734.00	9333.00	0	2.40	18.30	734.00
daysonmarket_median	13843.0	49.15	56.06	0.00	26.00	36.00	51.00	1987.00	0	**8.75**	168.60	25.00
mileage_median	13842.0	51463.27	51845.46	0.00	25.00	33596.50	96915.75	238541.00	1	0.68	-0.77	96890.75
year_median	13843.0	2015.95	4.72	1960.00	2013.00	2017.00	2020.00	2021.00	0	-2.25	14.09	7.00
vehicle_age_median	13842.0	4.08	4.63	0.00	0.00	3.00	7.00	60.00	1	2.32	14.98	7.00
owner_count_median	13804.0	1.43	0.62	1.00	1.00	1.00	2.00	6.00	39	1.30	1.35	1.00
horsepower_median	13837.0	240.90	56.99	78.00	188.00	243.00	283.00	630.00	6	0.85	2.05	95.00
engine_displacement_median	10450.0	2925.08	808.08	1200.00	2400.00	2500.00	3500.00	6800.00	**3393**	1.29	2.77	1100.00
city_fuel_economy_median	13827.0	20.65	3.96	10.00	18.00	20.00	22.00	126.00	16	**8.10**	185.87	4.00
highway_fuel_economy_median	13827.0	27.88	3.98	15.00	25.00	27.50	30.00	105.00	16	**3.88**	66.79	5.00
back_legroom_median	13824.0	37.71	1.57	0.00	36.80	37.80	38.60	47.50	19	**-3.53**	77.81	1.80
front_legroom_median	13836.0	41.89	0.77	36.60	41.30	41.80	42.20	67.00	7	**3.49**	88.62	0.90
fuel_tank_volume_median	13836.0	18.35	2.93	7.15	16.40	18.25	19.20	52.00	7	1.83	7.33	2.80
height_median	13837.0	65.15	4.95	46.85	61.60	66.00	68.00	107.50	6	0.18	0.91	6.40
length_median	13837.0	191.10	10.25	151.00	184.60	189.80	192.70	282.30	6	2.18	7.59	8.10
width_median	13837.0	77.02	4.97	65.70	72.90	75.55	80.00	105.90	6	1.06	1.56	7.10
pct_isCab	13816.0	0.16	0.16	0.00	0.05	0.12	0.21	1.00	27	2.04	5.91	0.17
pct_is_new	13843.0	0.27	0.32	0.00	0.00	0.00	0.59	1.00	0	0.55	-1.38	0.59
pct_salvage	13816.0	0.01	0.06	0.00	0.00	0.00	0.00	1.00	27	**10.34**	119.82	**0.00**
pct_theft_title	13816.0	0.00	0.02	0.00	0.00	0.00	0.00	0.76	27	**14.11**	427.34	**0.00**
n_listing_days	13843.0	46.32	33.72	1.00	18.00	39.00	68.00	272.00	0	0.98	0.95	50.00
listing_span_days	13843.0	405.59	282.37	0.00	223.00	362.00	508.00	3553.00	0	2.15	9.00	285.00

Bold values mark severe skewness ($|\text{Skew}| \geq 3$), substantial missingness ($\geq 1\%$), and zero IQR.

1.2.2 Distribution via histograms

Figure 1.1 illustrates the distributions of core attributes covering seller scale, pricing, turnover, and inventory age. Operational scale (`n_listings`), pricing (`price_median`), and turnover time (`daysonmarket_median`) show heavily right-skewed distributions. The majority of sellers list fewer than 500 vehicles and clear inventory within 200 days, whereas the right tail extends to 4,959 listings and 1,987 days.

In addition, the inventory distribution also shows a clear separation between two distinct dealership models: used-car dealers (`pct_is_new` $= 0$, high mileage) and new-car dealers (`pct_is_new` > 0, near-zero mileage). In contrast, `year_median` is strongly left-skewed, concentrating mainly in 2013–2020. Besides, among the remaining attributes, `owner_count_median` concentrates mostly at one or two previous owners, while vehicle specifications (`height_median`, `length_median`, `front_legroom_median`, `fuel_tank_volume_median`) form high-kurtosis peaks around standard passenger-vehicle dimensions.

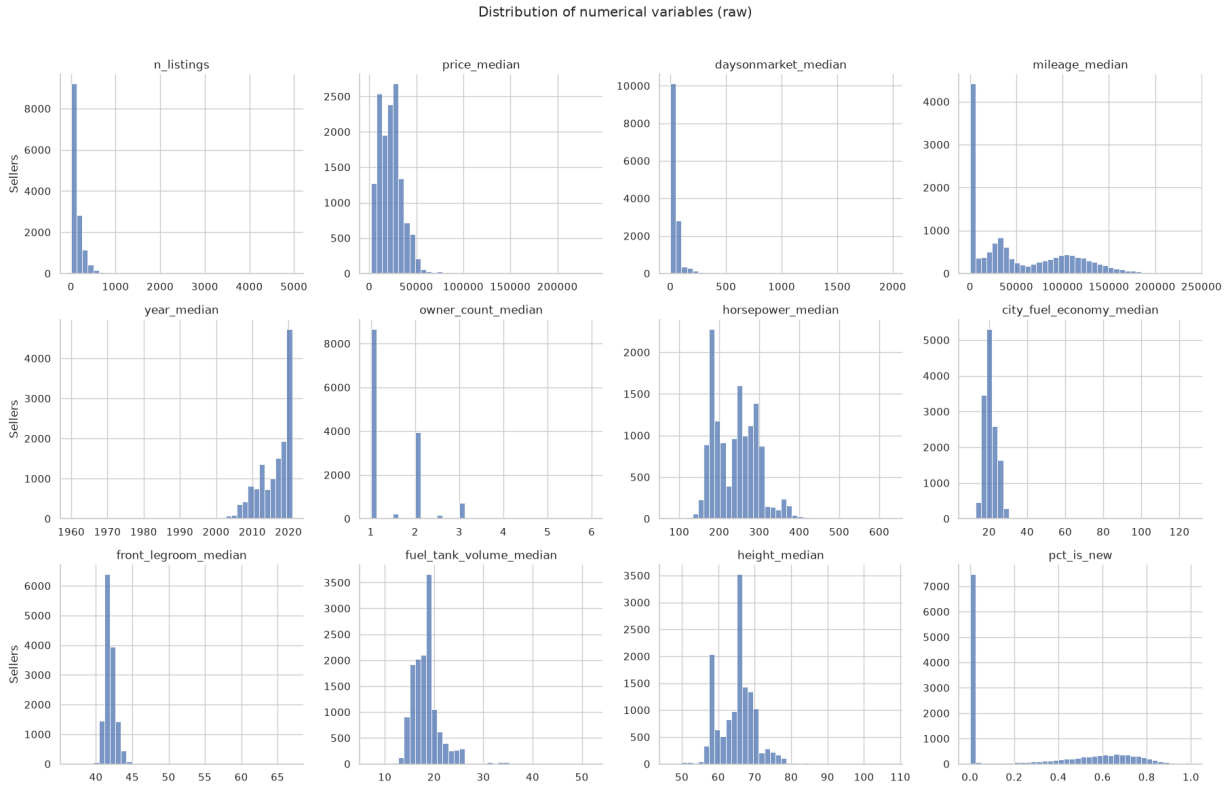


Figure 1.1: Histograms of the selected numerical variables.

1.2.3 Distribution of categorical features

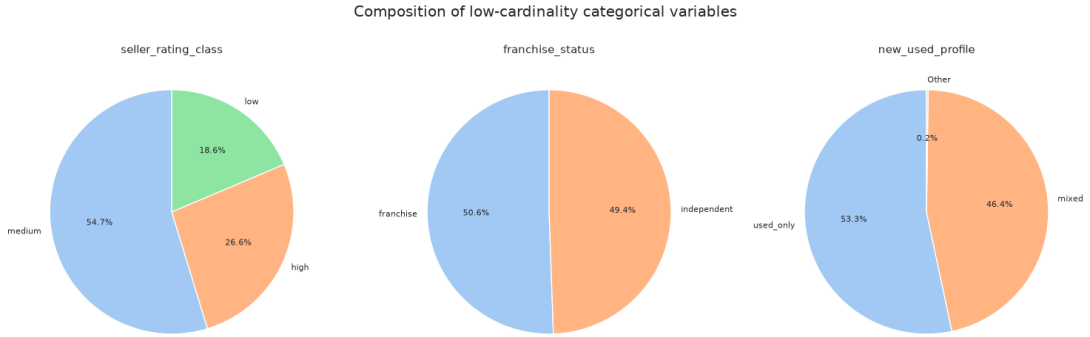


Figure 1.2: Composition of the low-cardinality categorical variables.

The pie charts (Figure 1.2) show that the target `seller_rating_class` is moderately imbalanced, led by *medium* (54.7%), followed by *high* (26.6%) and *low* (18.6%). In contrast, `franchise_status` and `new_used_profile` are evenly split between their values.

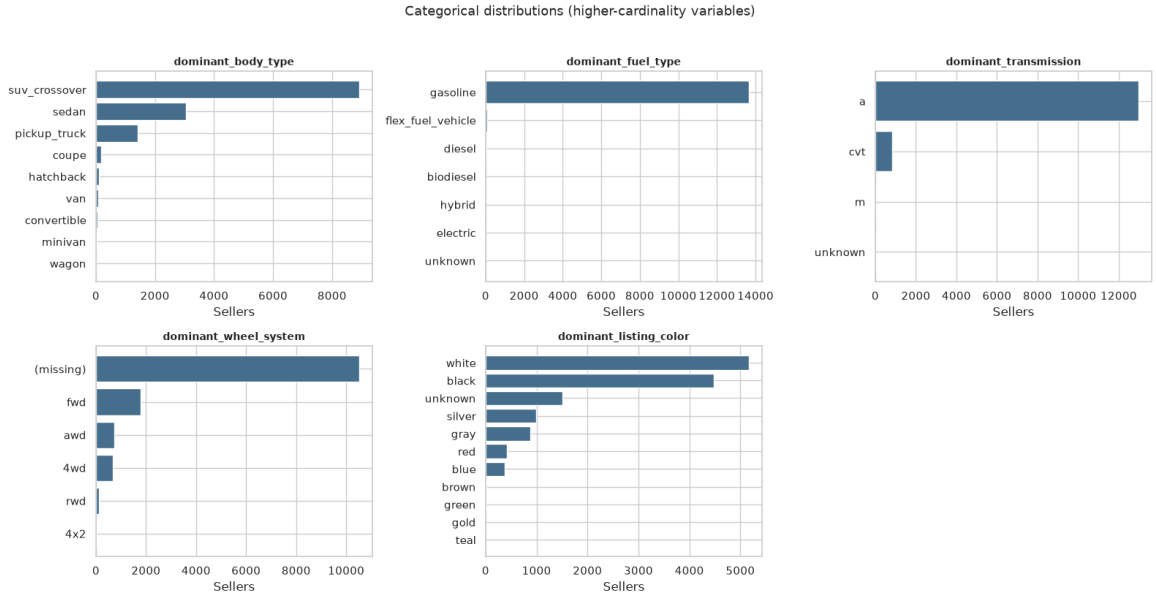


Figure 1.3: Distributions of the higher-cardinality categorical variables.

Among the higher-cardinality features, the bar charts (Figure 1.3) illustrate that automatic (`a`) dominates the `dominant_transmission` category, while `dominant_fuel_type` is concentrated in *gasoline*. In addition, `dominant_listing_color` is led by neutral colours (white, black, silver, grey), alongside a substantial *unknown* group (10.91%), while `dominant_wheel_system` is predominantly missing (75.98%). Lastly, SUVs cover a large proportion (61.1%) of `dominant_body_type`, followed by sedans (21.8%) and pickup trucks (12.1%), leaving all other body types below 5% in total.

1.3 Assessment of data quality

Our syntactic and semantic audit verified that there are no duplicated rows, specifically among the sellers' identifiers (`sp_id`). Secondly, there are no negative values in any numerical attribute. Thirdly, all percentage attributes (`pct_isCab`, `pct_is_new`, `pct_salvage`, `pct_theft_title`) are correctly bounded within the closed interval $[0, 1]$, and advertised discounts never exceed the original prices (`savings_amount_median` $<$ `price_median`). Moreover, temporal consistency holds across all records (`n_listing_days` $\leq$ `n_listings` and `n_listing_days` $\leq$ `listing_span_days` $+1$).

In addition, count and calendar-year medians (`year_median`, `owner_count_median`, `daysonmarket_median`) contain fractional `.5` values, which are valid because sellers with an even number of vehicle listings produce them.

We only made a simple change to the column `sp_id`, originally stored as `float64`, converting it to string.

1.3.1 Missing value handling

There are 20 attributes containing missing values: 17 with explicit `NaN` values and 3 with implicit *unknown* values (see Table 1.3 and Figure 1.4).

Table 1.3: Summary of explicit (`NaN`) and implicit (*unknown*) missing values. Bold values mark attributes with more than 1% missing.

Variable	Missing	Missing (%)	Type
`dominant_wheel_system`	**10,518**	**75.98%**	NaN
`engine_displacement_median`	**3,393**	**24.51%**	NaN
`dominant_listing_color`	**1,510**	**10.91%**	*unknown*
`dominant_fuel_type`, `dominant_transmission`	1–6	0.01–0.04%	*unknown*
`vehicle_age_median`	1	0.01%	NaN
14 minor numerical attributes*	1–39	0.01–0.28%	NaN

*`owner_count_median`, `pct_isCab`, `pct_salvage`, `pct_theft_title`, the physical dimensions, legrooms, fuel-economy attributes, `horsepower_median`, and `mileage_median`.

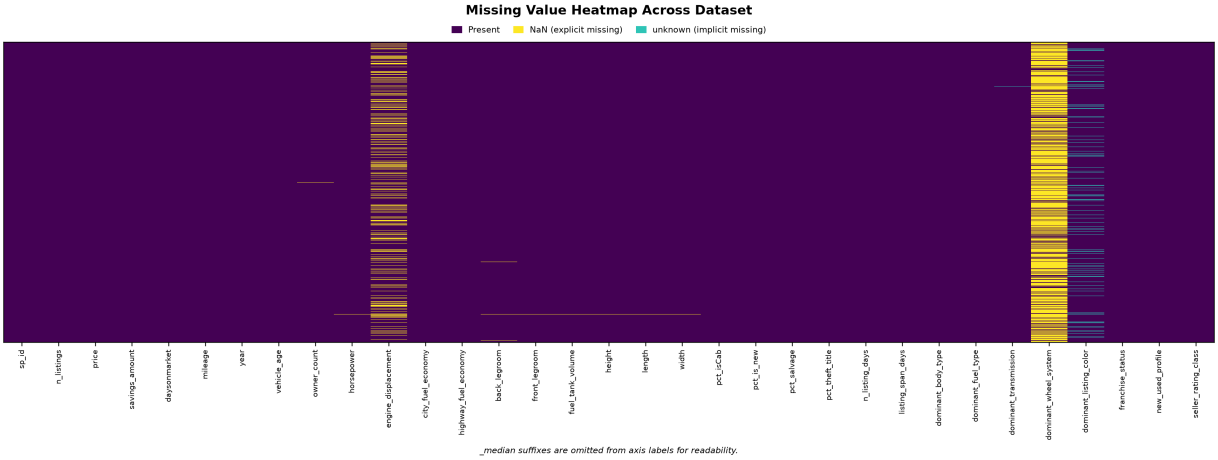


Figure 1.4: Missing-value heatmap: explicit `NaN` and implicit `unknown` values across the dataset.

We applied tailored strategies to handle missing values in each attribute as follows:

- `dominant_wheel_system`: dropped due to a high percentage of missing values (75.98%).
- `engine_displacement_median`: missing values (24.51%) are filled using the median grouped by `dominant_body_type` and `horsepower_median` quintiles, which ensures that the engine size reflects both the vehicle's body type and power.
- `dominant_listing_color`: implicit *unknown* values (10.91%) are kept as a separate category, preventing mode substitution from overly skewing the prevalence of the *white* label.
- `dominant_fuel_type` and `dominant_transmission`: missing values under the implicit *unknown* labels (1 and 6 instances) are converted to the group mode calculated across each `dominant_body_type`.
- `vehicle_age_median`: the single missing entry (0.01%) was computed using the formula $2020 - \texttt{year_median}$.
- The remaining 14 numerical attributes: minor missing values ($< 0.30\%$ each) are imputed using the group median calculated across each `dominant_body_type`, preserving expected characteristics across body types.

1.3.2 Outlier and noise detection

Inspection of the feature distributions (Figure 1.5) distinguishes true measurement noise from valid domain outliers:

True semantic noise. We identified 3 instances with `back_legroom_median` = 0 although their `dominant_body_type` is `suv_crossover` or `sedan`. Since this is physically impossible, the values were converted to the group median by body type, as in Section 1.3.1.

Valid domain outliers. The `pct_salvage` and `pct_theft_title` columns, which recorded 90.83% and 84.56% zeros, yield IQR = 0. These values represent valid second-hand or salvage dealerships rather than data errors, so we retained all records. Secondly, we examine extreme values across all attributes:

- `n_listings` (up to 4,959) is verified as a mega-dealership.
- `price_median` (up to \$236,438), `horsepower_median` (up to 630 HP) and `vehicle_age_median` (up to 60 years) are luxury, sport or classic-car dealers.
- `city_fuel_economy_median` (up to 126 MPG) represents electric-car dealers.
- `daysonmarket_median` (up to 1,987 days) reflects real market heterogeneity, as some dealers have slow-moving inventory.

Therefore, all records from these valid sellers were kept to retain niche market segments. In the end, 13,843 records were retained in the cleaned dataset.

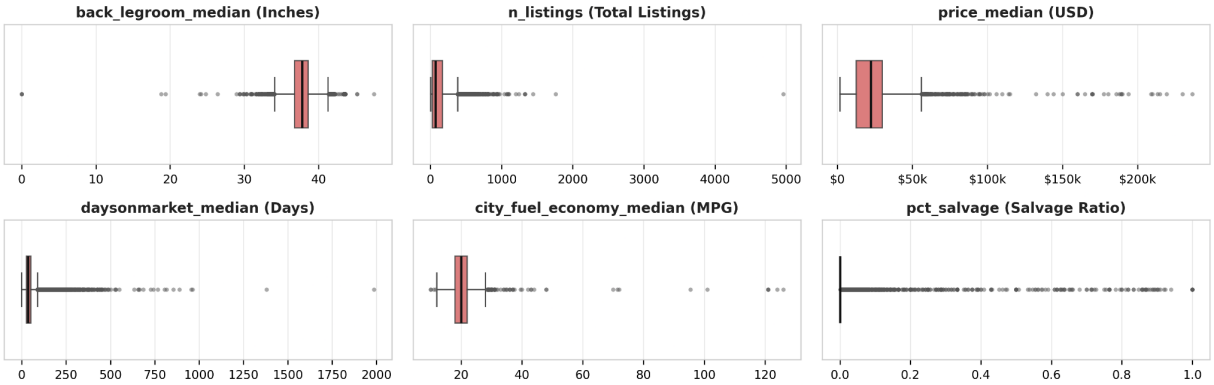


Figure 1.5: Boxplots of selected numerical attributes for outlier detection.

1.3.3 Variables transformation

To address the heavy right-skewness across the numerical attributes in the dataset, we applied a logarithmic transformation, $\log(1 + x)$, to the heavily right-skewed continuous features: `price_median`, `n_listings`, `daysonmarket_median`, `savings_amount_median`, `vehicle_age_median`, `city_fuel_economy_median`, `highway_fuel_economy_median`, `engine_displacement_median` and `fuel_tank_volume_median`. This compresses extreme right tails and restores near-symmetric distributions without undefined values at zero. We then performed feature scaling to ensure uniformity. We generated two standardised representations: Z-score standardisation to mitigate outlier leverage in distance-based algorithms, and Min-Max scaling for bounded metrics.

1.3.4 Pairwise correlations and variable elimination

Inter-feature dependencies were evaluated across the numerical attributes using the pairwise scatter-plot matrix (Figure 1.6) and Spearman's correlation matrix (Figure 1.7). We selected the Spearman method over Pearson because Figure 1.6 reveals strong non-linear monotonic patterns: specifically, `price_median` drops sharply as `mileage_median` and `vehicle_age_median` increase before flattening out, forming an L-shaped curve alongside the heavy right-tail outliers.

The correlation matrix also shows some important findings. Firstly, regarding highly correlated pairs, we retained both `mileage_median` and `vehicle_age_median` (0.93) as well as `n_listings` and `n_listing_days` (0.96) in the master dataset, because their relative ratios capture annual vehicle usage and active posting intensity. Secondly, within the technical specifications, `engine_displacement_median` correlates positively with `fuel_tank_volume_median` (0.79) and `horsepower_median` (0.77), but negatively with `city_fuel_economy_median` (-0.76), reflecting the trade-off between engine capacity and fuel efficiency.

Finally, we removed `year_median` (-0.98 with `vehicle_age_median`, since the latter equals 2020 $-$ `year_median`) and `highway_fuel_economy_median` (0.94 with `city_fuel_economy_median`, as they describe identical fuel efficiency), resulting in a final prepared dataset of 13,843 rows and 29 attributes.

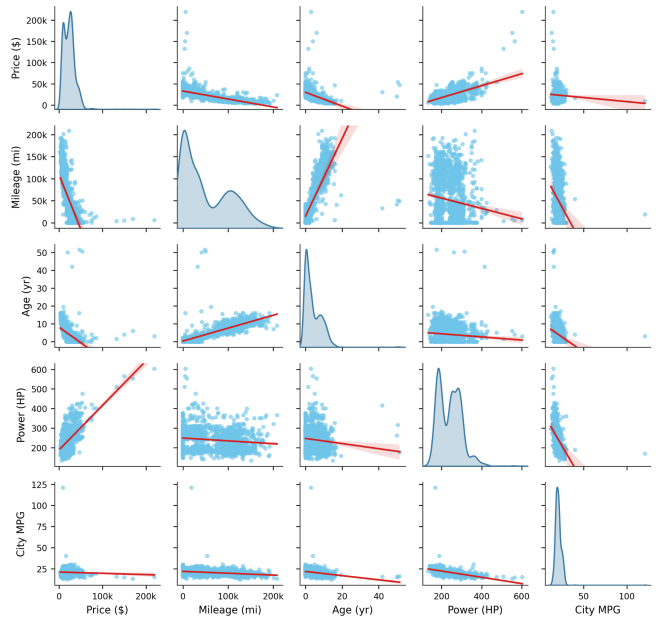


Figure 1.6: Pairwise scatter-plot matrix of the key numerical attributes.

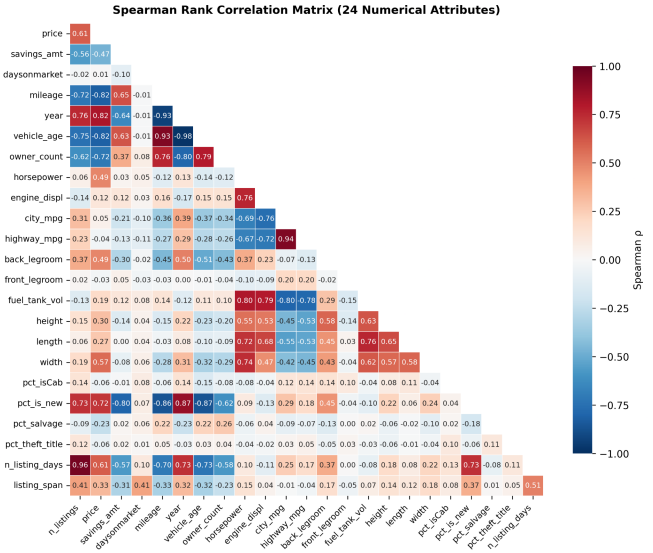


Figure 1.7: Spearman correlation matrix of the numerical attributes.

2 Clustering

(TODO: describe the clustering setup, the features used, and how dimensionality reduction was applied.)

2.1 Centroid-based

(TODO: K-Means; elbow and silhouette for choosing K.)

2.2 Density-based

(TODO: DBSCAN; k-NN distance plot to choose ε and MinPts.)

2.3 Hierarchical

(TODO: agglomerative clustering; linkage comparison and dendrograms.)

2.4 Final discussion

(TODO: compare the methods and state which worked best.)

3 Classification

The target variable is `seller_rating_class`. Describe the preprocessing, the train/test split, and the evaluation metrics (accuracy, precision, recall, F1, ROC-AUC).

3.1 KNN

(TODO: hyperparameter search results and per-class metrics.)

3.2 Naïve Bayes

(TODO: Gaussian and/or Categorical Naive Bayes results.)

3.3 Decision Trees

(TODO: tuning, pruning, and the resulting tree.)

3.4 Final conclusion

(TODO: compare all classifiers and recommend one.)

Table 3.1: Model comparison. *(Fill in.)*

Model	Accuracy	Macro-F1	Macro-AUC
k-NN			
Naïve Bayes			
Decision Tree			

4 Regression

(TODO: describe the regression target and predictors.)

4.1 Univariate regression

(TODO: simple regression per predictor.)

4.2 Multivariate regression

(TODO: multiple and multi-target regression.)

5 Pattern Mining

(TODO: describe the discretization/binning of numeric attributes and the transactional representation.)

5.1 Frequent patterns

(TODO: Apriori vs FP-Growth, support and itemset-size analysis.)

5.2 Association rules

(TODO: top rules, support/confidence/lift, and interpretation.)

6 Conclusion

(TODO: summarise the findings and the main takeaways.)

Team Contributions

Member	Contribution
Muhammad Iqbal Zain	
Truong Giang Do	
Vladislav Zhelezov	